Coordinate Geometry

ClipSAT Printable Practice Suite (clipsat.org) • Topic 10.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

In the diagram below, two chords, AB and CD, intersect at point P inside a circle. If $AP = 5$ cm, $PB = 8$ cm, and $CP = 4$ cm, what is the length of PD in cm?

According to the Power of a Point theorem for intersecting chords, the product of the segments of one chord is equal to the product of the segments of the other chord. In this case, $AP \cdot PB = CP \cdot PD$. We are given $AP = 5$ cm, $PB = 8$ cm, and $CP = 4$ cm. Substituting these values into the equation, we get $5 \cdot 8 = 4 \cdot PD$. This simplifies to $40 = 4 \cdot PD$. Dividing both sides by 4, we find $PD = 10$ cm. Therefore, the length of PD is 10 cm.

2 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

In the figure below with $AB \parallel CD$, $\text{Area}(\Delta ABE) = 36$ and $\text{Area}(\Delta CDE) = 81$. If base $AB = 10$, what is the length of base CD?

Because $\Delta ABE \sim \Delta CDE$, the ratio of their areas equals the square of the linear scale factor: $\frac{\text{Area}(\Delta CDE)}{\text{Area}(\Delta ABE)} = \frac{81}{36} = \frac{9}{4} = k^2$. Taking the square root gives $k = \sqrt{\frac{9}{4}} = \frac{3}{2}$. Therefore, $CD = k \cdot AB = \frac{3}{2} \times 10 = 15$.

3 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

The triangle ABC is translated on the coordinate plane according to the rule $(x, y) \to (x - 3, y + 2)$. If the vertices of the image triangle A'B'C' are A'(-1, 5), B'(3, 1), and C'(-2, -3), what are the coordinates of the original vertices A, B, and C?

The translation rule $(x, y) \to (x - 3, y + 2)$ means that to find the image coordinates (x', y') from the pre-image coordinates (x, y), we subtract 3 from the x-coordinate and add 2 to the y-coordinate: $x' = x - 3$ and $y' = y + 2$. To find the pre-image coordinates from the image coordinates, we reverse this process. This means we add 3 to the x-coordinate of the image and subtract 2 from the y-coordinate of the image: $x = x' + 3$ and $y = y' - 2$. For vertex A': $x = -1 + 3 = 2$, $y = 5 - 2 = 3$..

4 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

In ΔABC, $AD = 3$ and $DB = 9$ (so $AB = 12$); $AE = 4$ and $EC = 5$ (so $AC = 9$). If base $BC = 21$, what is the length of segment DE?

Compare the side ratios across the shared angle $\angle A$: $\frac{AD}{AC} = \frac{3}{9} = \frac{1}{3}$ and $\frac{AE}{AB} = \frac{4}{12} = \frac{1}{3}$. Because two pairs of corresponding sides are in the same constant ratio $1/3$ and include the shared angle $\angle A$, $\Delta ADE \sim \Delta ACB$ by SAS similarity in inverted order. Thus, $\frac{DE}{BC} = \frac{1}{3} \Rightarrow DE = 21 \times \frac{1}{3} = 7$.

© ClipSAT — clipsat.org | All Rights Reserved

5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

The diagonals of trapezoid $ABCD$ with $AB \parallel CD$ intersect at P. If $\text{Area}(\Delta APB) = 16$ and $\text{Area}(\Delta CPD) = 49$, what is the total area of trapezoid $ABCD$?

Because $\Delta APB \sim \Delta CPD$, the ratio of areas is $16/49 = (4/7)^2$. For any trapezoid, the side triangles have equal area given by the geometric mean of the base triangles: $\text{Area}(\Delta APD) = \text{Area}(\Delta BPC) = \sqrt{16 \times 49} = 4 \times 7 = 28$. The total area of the trapezoid is $16 + 49 + 28 + 28 = 121$ (which equals $(4 + 7)^2 = 11^2 = 121$).

6 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In ΔABC, segment $DE \parallel BC$ with D on AB and E on AC. If $AD = 5$, $DB = 10$, and $DE = 7$, what is the length of base BC?

Because $DE \parallel BC$, $\Delta ADE \sim \Delta ABC$ by AA. The parallel bases correspond to full triangle sides from vertex A: $AB = AD + DB = 5 + 10 = 15$. Setting up the whole-to-part ratio: $\frac{DE}{BC} = \frac{AD}{AB} \Rightarrow \frac{7}{BC} = \frac{5}{15} = \frac{1}{3} \Rightarrow BC = 7 \times 3 = 21$. Note: Using part-to-part $5/10$ gives the distractor 14.

7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In the hourglass figure below, segment $AB \parallel CD$, and segments AC and BD intersect at E. If $AB = 8$, $CD = 12$, and $AE = 6$, what is the length of segment CE?

Because $AB \parallel CD$, alternate interior angles $\angle A \cong \angle C$ and $\angle B \cong \angle D$ combined with vertical angles at E prove that $\Delta ABE \sim \Delta CDE$. Setting up the ratio of corresponding sides: $\frac{AE}{CE} = \frac{AB}{CD} \Rightarrow \frac{6}{CE} = \frac{8}{12} = \frac{2}{3} \Rightarrow CE = \frac{6 \times 3}{2} = 9$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A right triangle has legs of lengths 15 and 20. What is the exact length of the altitude drawn perpendicular to the hypotenuse?

First compute the hypotenuse using the Pythagorean theorem: $c = \sqrt{15^2 + 20^2} = \sqrt{225 + 400} = \sqrt{625} = 25$. Equating the area calculated in two ways: $\text{Area} = \frac{1}{2}ab = \frac{1}{2}ch \Rightarrow ab = ch$. Substituting: $15 \times 20 = 25 \cdot h \Rightarrow 300 = 25h \Rightarrow h = \frac{300}{25} = 12$.

© ClipSAT — clipsat.org | All Rights Reserved

9 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In ΔABC, $\angle ADE \cong \angle C$ with $AD = 5$ and $AC = 15$. If the total area of ΔABC is 108, what is the area of quadrilateral $BDEC$?

Because $\Delta ADE \sim \Delta ACB$, the linear ratio is $k = \frac{AD}{AC} = \frac{5}{15} = \frac{1}{3}$. The area ratio is $k^2 = (1/3)^2 = 1/9$. Thus, $\text{Area}(\Delta ADE) = \frac{1}{9} \times 108 = 12$. The area of quadrilateral $BDEC = \text{Area}(\Delta ABC) - \text{Area}(\Delta ADE) = 108 - 12 = 96$.

10 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In ΔABC shown below, segment $DE \parallel BC$. If $AD = 6$, $DB = 9$, and $AE = 8$, what is the length of segment EC?

By the Triangle Proportionality Theorem (Thales' Theorem), a line parallel to one side of a triangle divides the other two sides proportionally: $\frac{AD}{DB} = \frac{AE}{EC}$. Substituting the given values:
$\frac{6}{9} = \frac{8}{EC} \Rightarrow \frac{2}{3} = \frac{8}{EC} \Rightarrow EC = \frac{3 \times 8}{2} = 12$.

11 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In ΔABC, $DE \parallel BC$ with D on AB and E on AC. If $AD = 4$ and $AB = 10$, what fraction of the area of ΔABC is occupied by trapezoid $BDEC$?

Since $DE \parallel BC$, $\Delta ADE \sim \Delta ABC$ with linear scale factor $k = \frac{AD}{AB} = \frac{4}{10} = \frac{2}{5}$. The ratio of the area of ΔADE to the area of ΔABC is $k^2 = (2/5)^2 = 4/25$. The area of trapezoid $BDEC$ is the remaining fraction:
$1 - \frac{4}{25} = \frac{21}{25}$.

12 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Trapezoid $ABCD$ has parallel bases $AB = 6$ and $CD = 14$. The diagonals AC and BD intersect at point P. What is the ratio of $\text{Area}(\Delta APB)$ to $\text{Area}(\Delta CPD)$?

Because $AB \parallel CD$, alternate interior angles and vertical angles prove that $\Delta APB \sim \Delta CPD$. The linear scale factor is $k = \frac{AB}{CD} = \frac{6}{14} = \frac{3}{7}$. The ratio of their areas equals the square of the linear scale factor:
$k^2 = \left(\frac{3}{7}\right)^2 = \frac{9}{49}$.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In ΔABC, point D lies on AB and E on AC such that $\angle AED \cong \angle B$. If $AE = 6$, $AD = 8$, $AC = 18$, and base $BC = 27$, what is the length of segment DE?

Since $\angle A$ is shared and $\angle AED \cong \angle B$, $\Delta AED \sim \Delta ABC$ by AA similarity. The ratio of corresponding sides is $\frac{DE}{BC} = \frac{AD}{AC}$. Substituting the known lengths: $\frac{DE}{27} = \frac{8}{18} = \frac{4}{9} \Rightarrow DE = 27 \times \frac{4}{9} = 12$.

© ClipSAT — clipsat.org | All Rights Reserved

14 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

In trapezoid $ABCD$ with $AB \parallel CD$, $AB = 8$ and $CD = 12$. The diagonals intersect at P. If the entire diagonal $AC = 30$, what is the length of segment AP?

Because $\Delta APB \sim \Delta CPD$, corresponding segments along the diagonal satisfy $\frac{AP}{PC} = \frac{AB}{CD} = \frac{8}{12} = \frac{2}{3}$. The total length is $AC = AP + PC = 30$. Thus, $AP = 30 \times \frac{2}{2+3} = 30 \times \frac{2}{5} = 12$ (and $PC = 18$).

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)** • **Cross-Lesson: 5.01 Right Triangles & Trigonometry**

In the bowtie figure with $AB \parallel CD$, segments AC and BD intersect at E. If $AB = 10$, $CD = 15$, and the entire segment $BD = 35$, what is the length of segment BE?

Since $AB \parallel CD$, $\Delta ABE \sim \Delta CDE$. The ratio of corresponding sides is $\frac{BE}{ED} = \frac{AB}{CD} = \frac{10}{15} = \frac{2}{3}$. The total length $BD = BE + ED = 35$. Therefore, $BE = 35 \times \frac{2}{2+3} = 35 \times \frac{2}{5} = 14$.

16 **[Challenging]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)** • **Cross-Lesson: 6.01 Polygons & Quadrilaterals**

In ΔABC, ray AD bisects $\angle A$ with point D on side BC. If $AB = 8$, $AC = 12$, and $BD = 6$, what is the length of segment DC?

By the Angle Bisector Theorem, an angle bisector divides the opposite side into segments proportional to the adjacent sides: $\frac{BD}{DC} = \frac{AB}{AC}$. Substituting: $\frac{6}{DC} = \frac{8}{12} = \frac{2}{3} \Rightarrow 2 \cdot DC = 18 \Rightarrow DC = 9$.

17 **[Challenging]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)** • **Cross-Lesson: 7.01 Circles**

In ΔABC, ray AD bisects $\angle A$. If $AB = x$, $AC = x + 4$, $BD = 4$, and $DC = 6$, what is the length of side AC?

Apply the Angle Bisector Theorem: $\frac{BD}{DC} = \frac{AB}{AC} \Rightarrow \frac{4}{6} = \frac{x}{x+4} \Rightarrow \frac{2}{3} = \frac{x}{x+4}$. Cross-multiplying: $2(x + 4) = 3x \Rightarrow 2x + 8 = 3x \Rightarrow x = 8$. Side $AC = x + 4 = 8 + 4 = 12$.

18 **[Challenging]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)** • **Cross-Lesson: 8.01 Area, Surface Area & Volume**

In right ΔABC, altitude CD to hypotenuse AB measures 6, and the entire hypotenuse $AB = 13$. What is the length of the longer of the two segments into which D divides AB?

Let the two segments be p and q. We know $p + q = 13$ and $p \cdot q = CD^2 = 6^2 = 36$. These are the roots of the quadratic equation $t^2 - 13t + 36 = 0$. Factoring: $(t - 4)(t - 9) = 0$. The segments are 4 and 9. The longer segment is 9.

© ClipSAT — clipsat.org | All Rights Reserved

19 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 9.01 Transformations & Symmetry

In ΔABC, segment $DE \parallel BC$. If $AD = 8$, $DB = 4$, and $AC = 18$, what is the length of segment AE?

Total length $AB = AD + DB = 8 + 4 = 12$. The ratio of AD to AB is $\frac{AD}{AB} = \frac{8}{12} = \frac{2}{3}$. By similarity, $\frac{AE}{AC} = \frac{AD}{AB} = \frac{2}{3}$. Therefore, $AE = 18 \times \frac{2}{3} = 12$.

20 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 1.01 Foundations & Reasoning

In ΔABC, ray AD bisects $\angle A$ with D on BC. If $AB = 10$, $AC = 15$, and the entire base $BC = 20$, what is the length of segment BD?

By the Angle Bisector Theorem: $\frac{BD}{DC} = \frac{AB}{AC} = \frac{10}{15} = \frac{2}{3}$. The total base is $BC = BD + DC = 20$. Therefore, $BD = 20 \times \frac{2}{2+3} = 20 \times \frac{2}{5} = 8$ (and $DC = 12$).

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 6x + 6$ in review: circumference and area, evaluate $f(2)$.

a. 18 ✓

b. 24

c. 12

d. 22

Substitute $x = 2$: $f(2) = 6(2) + 6 = 18$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 2x + 7$ in arc length and sector area, evaluate $f(2)$.

a. 11 ✓

b. 13

c. 4

d. 15

Substitute $x = 2$: $f(2) = 2(2) + 7 = 11$.

© ClipSAT — clipsat.org | All Rights Reserved

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 3x + 8$ in extension: chords of a circle, evaluate $f(2)$.

a. 14 ✓

b. 17

c. 6

d. 18

Substitute $x = 2$: $f(2) = 3(2) + 8 = 14$.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for extension: chords of a circle consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean?

a. 21.5

b. 22.5

c. 23.5 ✓

d. 25.5

The sum is $17 + 21 + 25 + 31 = 94$, and dividing by 4 gives 23.5.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

In ΔABC studying angles from intersecting chords, secants, and tangents, $m\angle A = 43^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 85° ✓

b. 95°

c. 77°

d. 137°

Using the triangle angle-sum theorem, $m\angle C = 180^\circ - (43^\circ + 52^\circ) = 85^\circ$.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for tangents to a circle consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5 ✓

d. 19.5

The sum is $11 + 15 + 19 + 25 = 70$, and dividing by 4 gives 17.5.

© ClipSAT — clipsat.org | All Rights Reserved

27 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the midpoint of the segment connecting $(-2, -2)$ and $(-8, 7)$.

a. (-6, 9)

b. (-5, -2)

c. (-10, 5)

d. **(-5, 5/2)** ✓

Midpoint $= \left(\frac{-2+(-8)}{2}, \frac{-2+7}{2}\right) = (-5, 5/2)$.

28 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the distance between the points $(-6, 4)$ and $(4, -8)$.

a. $3\sqrt{5}$

b. $2\sqrt{61}$ ✓

c. 22

d. $2\sqrt{62}$

Distance $= \sqrt{(4-(-6))^2 + (-8-4)^2} = \sqrt{244} = 2\sqrt{61}$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Find the slope of the line through $(-8, 8)$ and $(-2, 7)$.

a. 1/6

b. **-1/6** ✓

c. -6

d. 6

Slope $= \frac{7-8}{-2-(-8)} = \frac{-1}{6} = -1/6$.

© ClipSAT — clipsat.org | All Rights Reserved

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In the coordinate plane, quadrilateral $ABCD$ has vertices $A(-2, 3)$, $B(4, 5)$, $C(6, -1)$, and $D(0, -3)$. Which of the following correctly classifies quadrilateral $ABCD$ and gives its exact area?

a. **A square with an area of** 40 ✓

b. A rectangle that is not a square, with an area of 40

c. A rhombus that is not a square, with an area of 40

d. A square with an area of 64

To classify quadrilateral $ABCD$ and determine its area, we examine its side lengths, slopes, and dimensions: 1. **Side Lengths (Distance Formula):** - $AB = \sqrt{(4-(-2))^2 + (5-3)^2} = \sqrt{6^2 + 2^2} = \sqrt{36+4} = \sqrt{40}$ - $BC = \sqrt{(6-4)^2 + (-1-5)^2} = \sqrt{2^2 + (-6)^2} = \sqrt{4+36} = \sqrt{40}$ - $CD = \sqrt{(0-6)^2 + (-3-(-1))^2} = \sqrt{(-6)^2 + (-2)^2} = \sqrt{36+4} = \sqrt{40}$ - $DA = \sqrt{(-2-0)^2 + (3-(-3))^2} = \sqrt{(-2)^2 + 6^2} = \sqrt{4+36} = \sqrt{40}$ Since.

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In the xy-plane, a parabola $\mathcal{P}$ is defined as the set of all points equidistant from the focus $F(1, 2)$ and the directrix line $y = -2$. The line $x - y = 1$ intersects $\mathcal{P}$ at two distinct points, A and B. What is the length of the segment AB?

a. $4\sqrt{2}$

b. 8

c. $8\sqrt{2}$ ✓

d. 16

Step 1: Derive the equation of the parabola $\mathcal{P}$ using the focus-directrix definition. By definition, any point $P(x, y)$ on the parabola satisfies: $\text{distance}(P, F) = \text{distance}(P, \text{directrix})$ The distance from (x, y) to $F(1, 2)$ is $\sqrt{(x-1)^2 + (y-2)^2}$, and the perpendicular distance to the directrix $y = -2$ is $|y-(-2)| = |y+2|$. Equating the squared distances: $(x-1)^2 + (y-2)^2 = (y+2)^2$ Expanding the terms involving y.

© ClipSAT — clipsat.org | All Rights Reserved

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Which of the following is an equation of the line that passes through the point $(4, -1)$ and is perpendicular to the line $y = 2x + 3$?

a. $y = -\frac{1}{2}x + 1$ ✓

b. $y = -\frac{1}{2}x - 3$

c. $y = 2x - 9$

d. $y = \frac{1}{2}x - 3$

To find the equation of the line perpendicular to $y = 2x + 3$, first determine the required slope. The given line has a slope of $m_1 = 2$. Two non-vertical lines are perpendicular if and only if the product of their slopes is -1 (i.e., their slopes are negative reciprocals): $m_2 = -\frac{1}{m_1} = -\frac{1}{2}$ Next, use the point-slope form $y - y_1 = m(x - x_1)$ with the point $(x_1, y_1) = (4, -1)$ and slope $m = -\frac{1}{2}$: $y - (-1) = -\frac{1}{2}(x - 4)$ $y + 1 = -\frac{1}{2}x + 2$.

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the equation (in slope-intercept form) of the line through $(-8, -8)$ with slope $1/2$.

a. **y=1/2x-4** ✓

b. y=1/2(x-(-8))

c. y=1/2x

d. y=-4x+1/2

Using point-slope form: $y - (-8) = 1/2(x - (-8))$, which simplifies to $y = 1/2x - 4$.

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A line has slope -3. What is the slope of a line perpendicular to it?

a. 1

b. **1/3** ✓

c. -3

d. 3

Perpendicular slopes are negative reciprocals: $-\frac{1}{-3} = 1/3$.

© ClipSAT — clipsat.org | All Rights Reserved

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

In a plane, two horizontal lines ℓ_1 and ℓ_2 are parallel, with ℓ_1 positioned above ℓ_2. A zig-zag transversal path connects point A on ℓ_1 to point D on ℓ_2 through intermediate vertices B and C, which lie strictly between the two parallel lines. - Segment AB extends downward and to the right from A, forming an acute angle of 65° with line ℓ_1. - Segment BC extends downward and to the left from B to C. - Segment CD extends downward and to the right from C, meeting line ℓ_2 at an acute angle of 25°. If the ray that bisects $\angle BCD$ is perpendicular to segment AB, what is the measure of $\angle ABC$?

a. 115°

b. 130°

c. 140° ✓

d. 150°

To solve this crook-angle transversal problem, we construct auxiliary horizontal lines parallel to ℓ_1 and ℓ_2 through vertices B and C. 1. **Determine the direction of the angle bisector of $\angle BCD$:** - Segment AB slopes downward and to the right at an angle of 65° below the horizontal (direction angle -65°). - Since the ray bisecting $\angle BCD$ points into the interior of $\angle BCD$ (to the right) and is perpendicular to segment AB, its inclination angle relative to the ho.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A quadrilateral in the coordinate plane has vertices at $(0, 0)$, $(3, 0)$, $(3, 5)$, and $(0, 9)$. What is the volume of the three-dimensional solid generated by rotating this region 360° about the y-axis?

a. 57π ✓

b. 66π

c. 72π

d. 81π

To find the volume of the solid generated by revolving the region about the y-axis, we can decompose the quadrilateral horizontally at $y = 5$ into two simpler geometric shapes: 1. **Bottom Region (from $y = 0$ to $y = 5$):** This region is a rectangle bounded by $x = 0$, $x = 3$, $y = 0$, and $y = 5$. Revolving this rectangle around the y-axis generates a right circular cylinder with radius $r = 3$ and height $h_1 = 5$. $V_{\text{cylinder}} = \pi r^2 h_1 = \pi(3^2)(5) = 45\pi$ 2. **Top Region (from $y = 5$.

© ClipSAT — clipsat.org | All Rights Reserved

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In the coordinate plane, point $P(7, -2)$ is rotated 90° counterclockwise about the point $C(3, 1)$ to produce image point P'. What are the coordinates of P'?

a. $(6, 5)$ ✓

b. $(3, 4)$

c. $(2, 7)$

d. $(0, -3)$

To rotate a point $P(x, y)$ by 90° counterclockwise about an arbitrary center $C(h, k)$: 1. Translate the center $C(3, 1)$ to the origin by finding the displacement vector from C to P: $\Delta x = x - h = 7 - 3 = 4$ $\Delta y = y - k = -2 - 1 = -3$ 2. Apply a 90° counterclockwise rotation about the origin, which maps $(\Delta x, \Delta y) \to (-\Delta y, \Delta x)$: $\Delta x' = -(-3) = 3$ $\Delta y' = 4$ 3. Translate back by adding the coordinates of center $C(3, 1)$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 8.01 Area, Surface Area & Volume

A right circular cone with radius R and height H is sliced by a plane parallel to its base at a distance of $\frac{1}{3}H$ from the apex. What is the area of the 2D cross-section formed by this plane?

a. $\frac{1}{9}\pi R^2$ ✓

b. $\frac{1}{3}\pi R^2$

c. $\frac{4}{9}\pi R^2$

d. $\frac{2}{3}\pi R^2$

The cross-section formed by a plane parallel to the base of a cone is a circle similar to the base. Let the radius of the cross-sectional circle be r and its height from the apex be h. The height of the original cone is H and its radius is R. By similar triangles, the ratio of the radius to the height is constant: $\frac{r}{h} = \frac{R}{H}$. In this problem, the plane is at a distance of $\frac{1}{3}H$ from the apex, so $h = \frac{1}{3}H$. Substituting this into the similarity ratio.

© ClipSAT — clipsat.org | All Rights Reserved

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 9.01 Transformations & Symmetry

In the coordinate plane, ΔABC has vertices $A(0, 0)$, $B(4, 0)$, and $C(2, 4)$. A dilation centered at the origin maps ΔABC to ΔDEF with vertices $D(0, 0)$, $E(10, 0)$, and $F(5, 10)$. If the area of ΔABC is 8, what is the area of ΔDEF?

a. **50** ✓

b. 20

c. 40

d. 64

The linear scale factor of the dilation is $k = \frac{OE}{OB} = \frac{10}{4} = 2.5$. Areas scale by k^2: $k^2 = (2.5)^2 = 6.25$. Therefore, $\text{Area}(\Delta DEF) = 8 \times 6.25 = 50$. Alternatively, base $DE = 10$ and height $h = 10$, so $\text{Area} = \frac{1}{2}(10)(10) = 50$.

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 1.01 Foundations & Reasoning

In ΔABC, side AB is congruent to side AC, and the measure of $\angle A$ is 50°. What is the measure of $\angle B$?

a. 50°

b. 65° ✓

c. 80°

d. 130°

According to the Isosceles Triangle Theorem, if two sides of a triangle are congruent, the angles opposite those sides are congruent. Since side AB is congruent to side AC, the angles opposite these sides are congruent: $m\angle B = m\angle C$. By the Triangle Angle-Sum Theorem, the sum of the interior angles in any triangle is 180°: $m\angle A + m\angle B + m\angle C = 180^\circ$ Substitute the given angle measure $m\angle A = 50^\circ$ and $m\angle C = m\angle B$: $50^\circ + 2(m\angle B) = 180^\circ$.

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 11.01 Probability

Triangle ABC has $\angle A = 46^\circ$ and $\angle B = 62^\circ$. A line through C parallel to AB is drawn, forming an exterior angle at C on the far side from the triangle. What is the measure of that exterior angle, and how does it relate to $\angle A$ and $\angle B$?

a. 72°

b. **108°** ✓

c. 46°

d. 62°

Since $\angle C = 180 - 46 - 62 = 72^\circ$, its exterior angle is $180 - 72 = 108^\circ$, which equals $\angle A + \angle B = 46 + 62 = 108^\circ$ by the Exterior Angle Theorem.

© ClipSAT — clipsat.org | All Rights Reserved

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 2.01 Lines & Angles

Pythagorean theorem applies to:

a. All triangles

b. Equilateral

c. **Right triangles** ✓

d. Obtuse

The Pythagorean theorem $a^2 + b^2 = c^2$ applies exclusively to right-angled triangles, where c is the hypotenuse opposite the 90° angle.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 3.01 Triangles & Congruence

A point is chosen at random inside a 19×19 square. A smaller 15×15 square sits in one corner. Find the probability the point lands inside the smaller square.

a. $\frac{15}{19}$

b. $\frac{225}{362}$

c. $\frac{136}{361}$

d. $\frac{225}{361}$ ✓

Geometric probability compares areas: $P = \frac{\text{favorable area}}{\text{total area}} = \frac{15^2}{19^2} = \frac{225}{361} = \frac{225}{361}$.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 4.01 Similarity & Proportional Reasoning

A regular 9-gon has each interior angle equal to $(4x + 28)^\circ$. Find the value of the interior angle.

a. 28°

b. 155°

c. **140°** ✓

d. 1260°

Sum of interior angles is 1260°, so each angle is $1260/9$. Setting $4x + 28 = 140/9$ and solving gives $x = 28$, so the angle $= 140^\circ$.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 5.01 Right Triangles & Trigonometry

Trapezoid parallel sides 5 and 9 height 4: area:

a. 14

b. 24

c. **28** ✓

d. 56

The area of a trapezoid is $A = \frac{b_1 + b_2}{2} \cdot h$. Substituting the given values: $A = \frac{5 + 9}{2} \cdot 4 = 7 \cdot 4 = 28$.

© ClipSAT — clipsat.org | All Rights Reserved